A Physics-Based Simulation Framework for Ion Transport and Surface Interaction Analysis in High-Aspect Ratio Etching

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Abstract

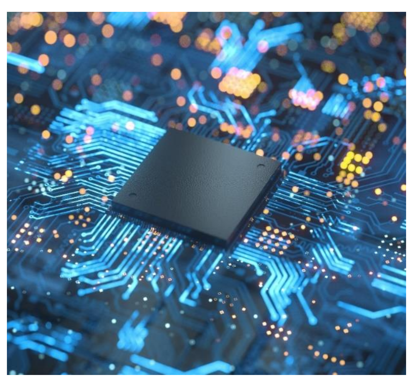
As the growth of the semiconductor industry accelerates, a precise understanding of ion-surface interactions within high aspect ratio (HAR) structures is essential. In particular, in the HAR etching process, complex physical behaviors such as ion scattering, energy transfer, and trajectory develop nonlinearly depending on the patterning characteristics within the structure, which is a key factor that makes accurate prediction and control of the etching process difficult.

In this study, molecular dynamics (MD) simulations were performed to theoretically elucidate the physical mechanisms of ion collision behavior in this process. The scattering characteristics and energy-angle distributions (EADs) for single bombardments were analyzed. Based on this data, a collision-based probability model was trained in a deep neural network (DNN) to build an AI-based encoding structure that can predict the physics-based probability distribution of ion-surface interactions.

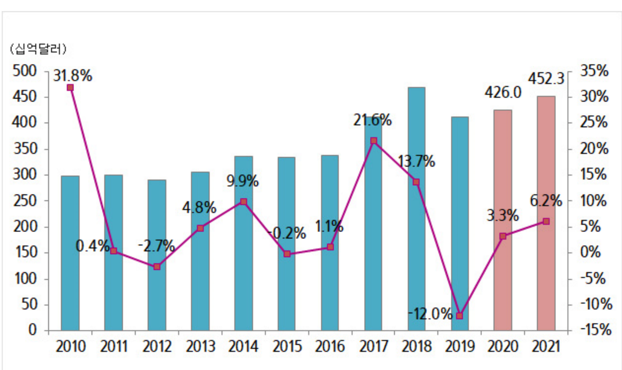
The probability distribution function predicted by the DNN is then applied to a monte carlo (MC) simulator, which is configured to dynamically track the ion trajectory, energy loss, sidewall collision frequency, and energy transfer as a function of position inside the trench. We propose a physics-based simulation framework to quantitatively analyze how ion transport pathways and energy transport properties change with structure under different HAR conditions.

Introduction

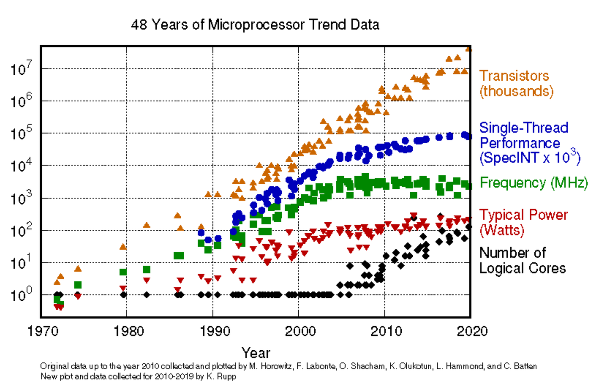
❑ Semiconductor Industry Trend



Semiconductor



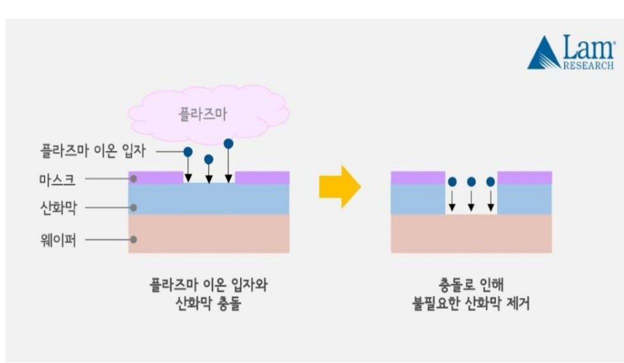
Market Grow Trend



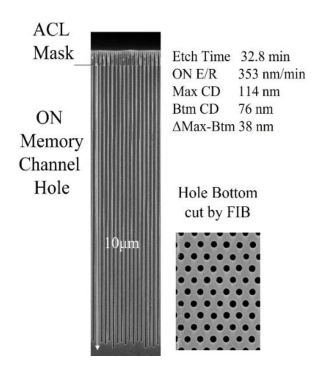
Moore's Law

❑ Etching Process

Semiconductor Fabrication

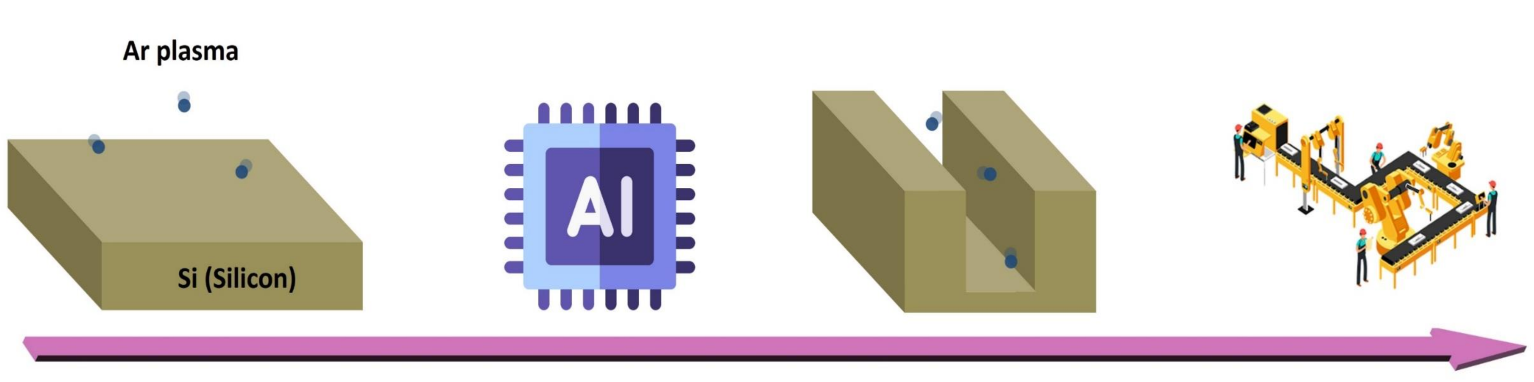


Plasma Etching Process



High Aspect Ratio

❑ Research Process

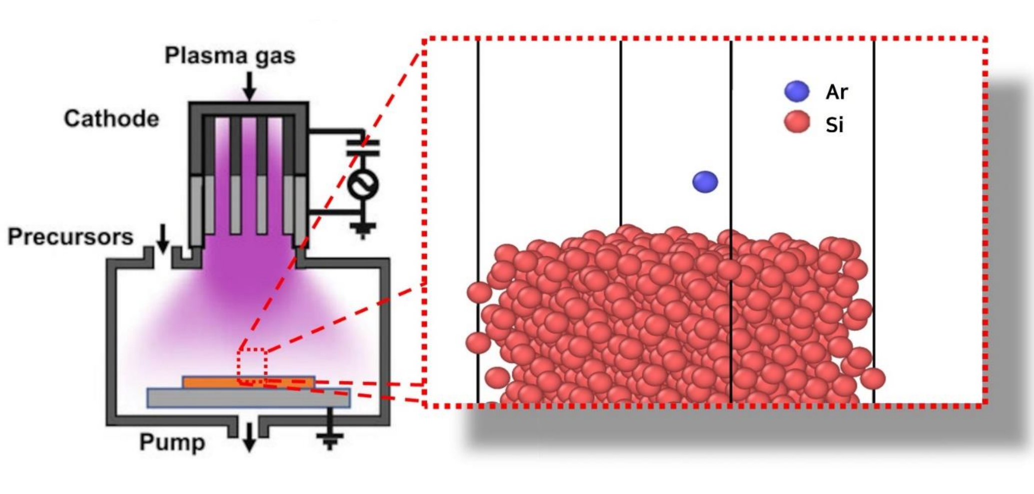


Etch MD Simulation — Mixture Density Network (MDN) — Monte Carlo Simulation — Process Improvement

Research Method

❑ Molecular Dynamics Simulation

- Performing MD simulations to obtain the energy-angle distribution following single bombardment of Ar

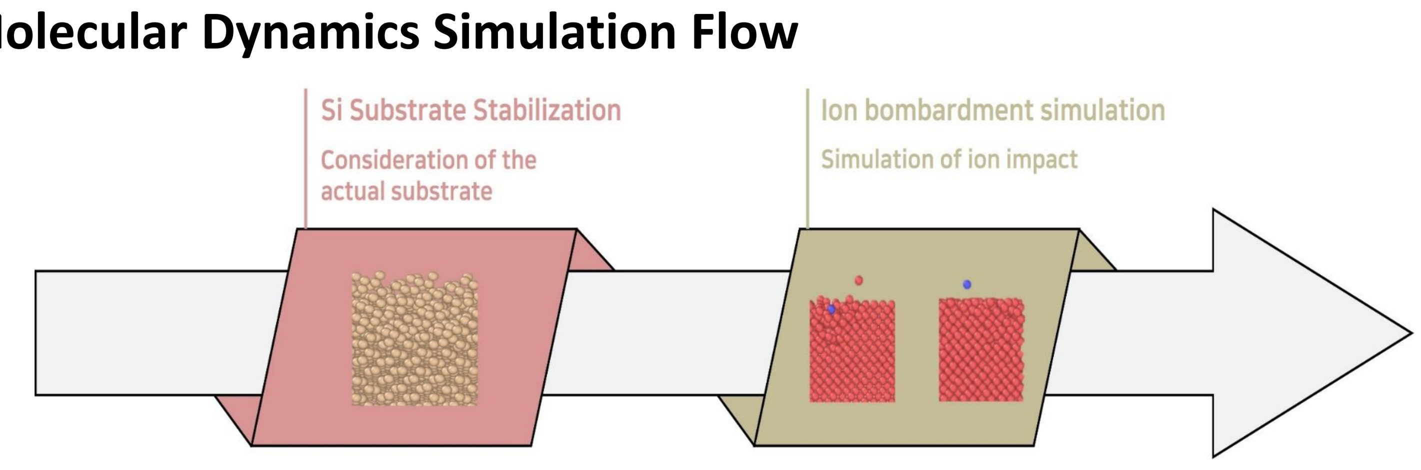


➢ Use Tools

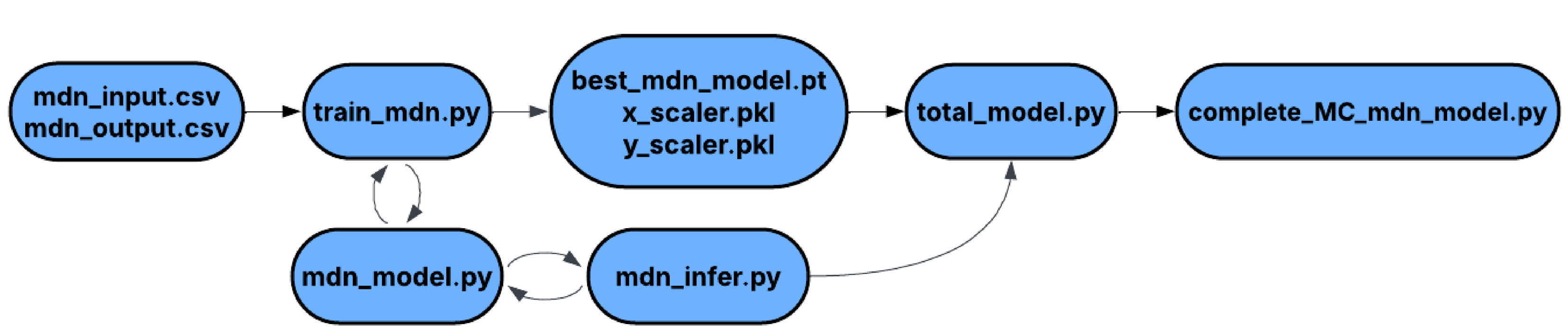
LAMMPS

OVITO

❑ Molecular Dynamics Simulation Flow



❑ Model Learning and Simulation Application Process



❑ Monte Carlo Simulation Modeling

Step 1. Setup - Node and 2D trench generation for energy accumulation tracking

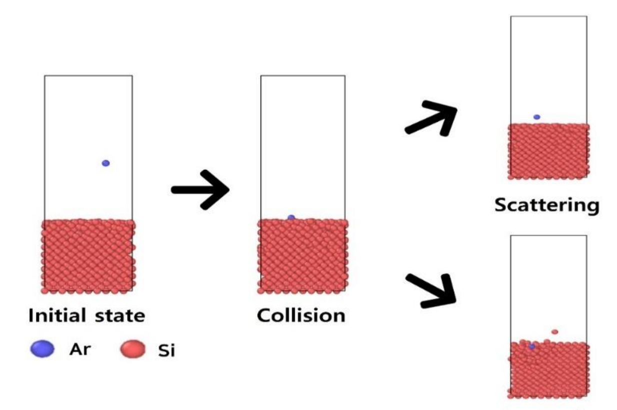
Step 2. Ion Simulation - Ions are randomly generated, and their trajectories are tracked

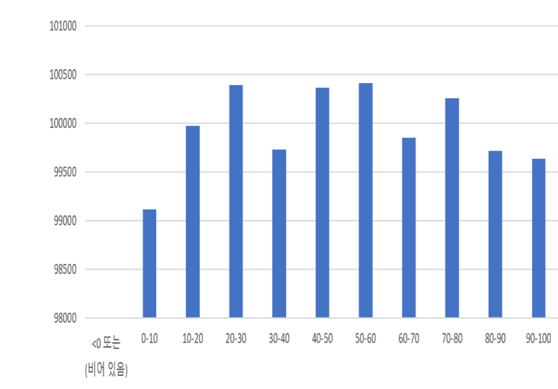
Step 3. MDN Utilization - When an ion collides with a trench wall, a pre-trained MDN model predicts the ion's reflection direction and energy

Step 4. Energy Recording - The energy transferred by the ion is recorded on the nodes surrounding the point of collision or implantation

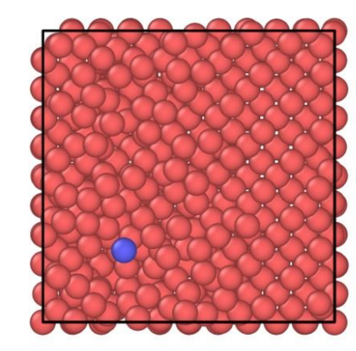
Research Result

❑ Simulation Visualization



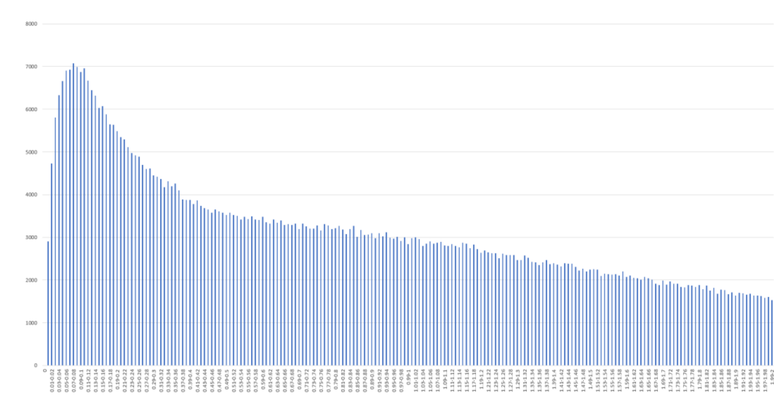


Initial Energy Distribution

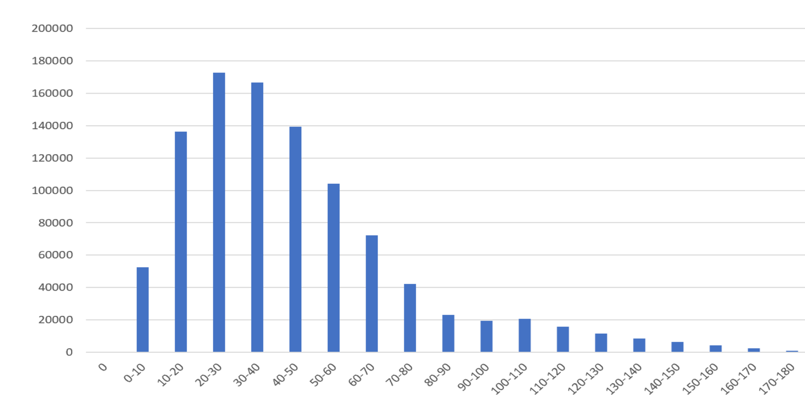


Injection Atom

❑ Energy & Angle Distribution

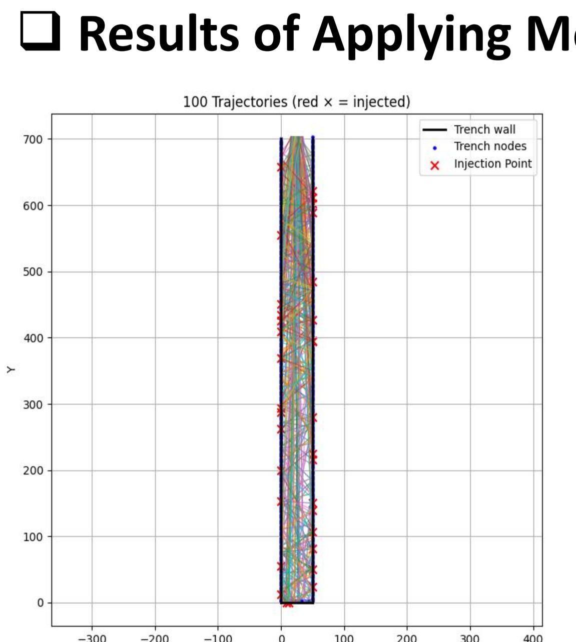


Energy Distribution

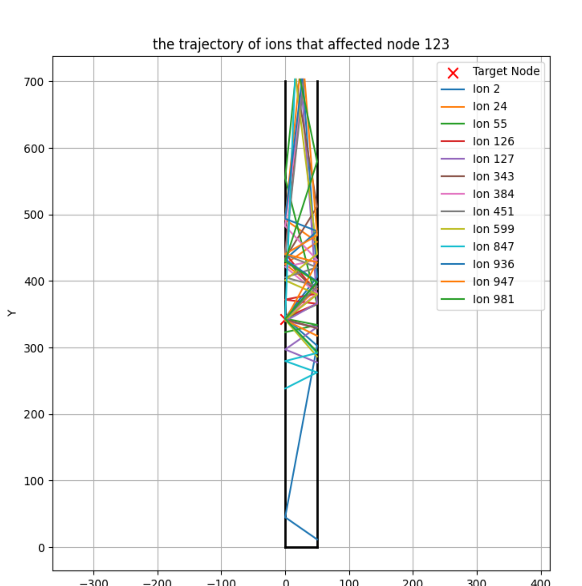


Angle Distribution

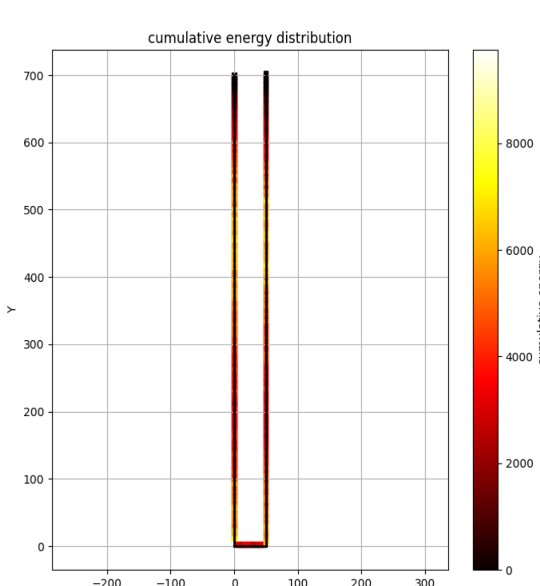
❑ Results of Applying Monte Carlo Simulation



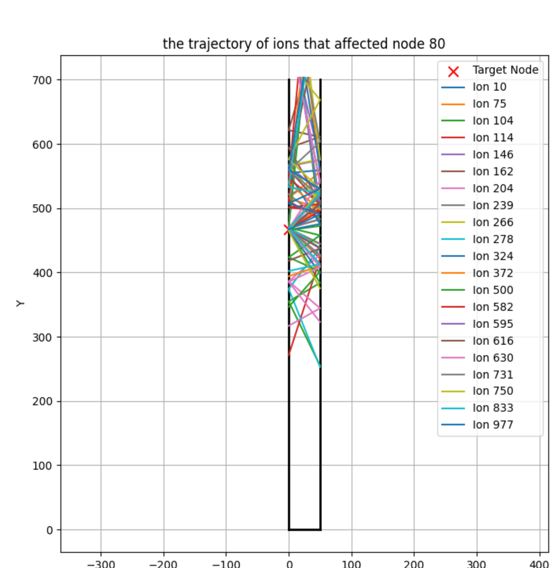
Trajectory of Ion



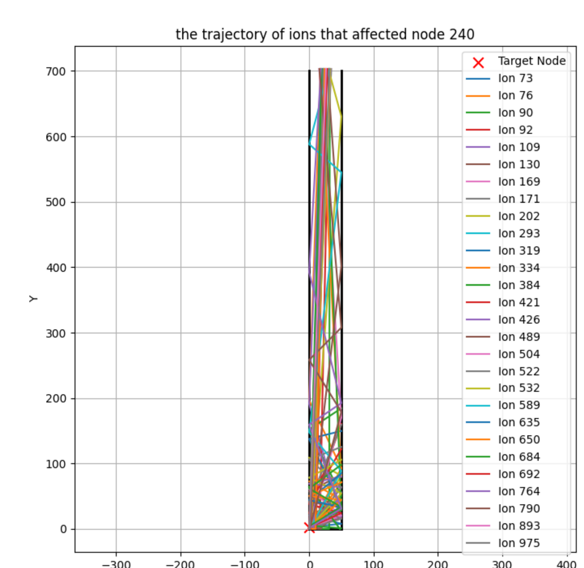
Trajectory of Reflect Ion



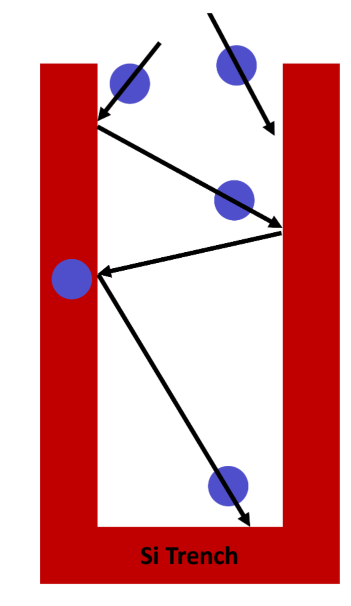
Energy accumulation



Ions colliding with nodes



Ion trajectories at a specific node



Visualization in Trench

Conclusion

- The simulator effectively models the stochastic properties of ion reflection and the implantation phenomenon, enabling more realistic etching process simulations than conventional methods.
- The simulator revealed a phenomenon in which the ion beam energy is unevenly distributed over the trench wall.
- Backtracking the trajectory of the ions that affected the energy concentration point can contribute to quantitative identification of the incident conditions, and to defining the initial conditions of the ideal ions, providing a blueprint for process optimization.
- This work demonstrates that incorporating AI surrogate models into traditional engineering simulations is a highly effective strategy for developing efficient and accurate prediction tools for next-generation semiconductor manufacturing.

Reference

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